

Artificial Intelligence Internship

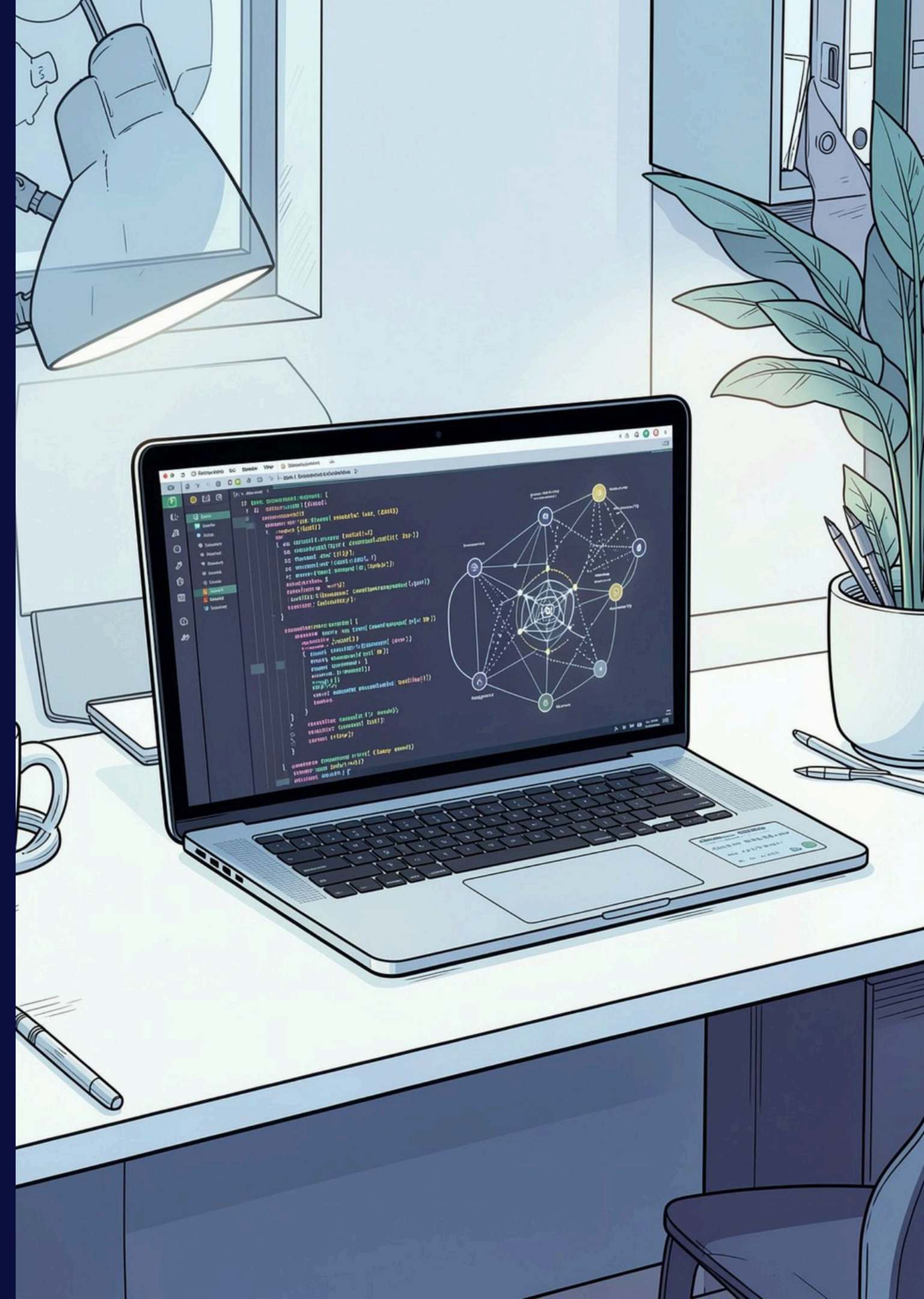
Internship Training and Learning Experience

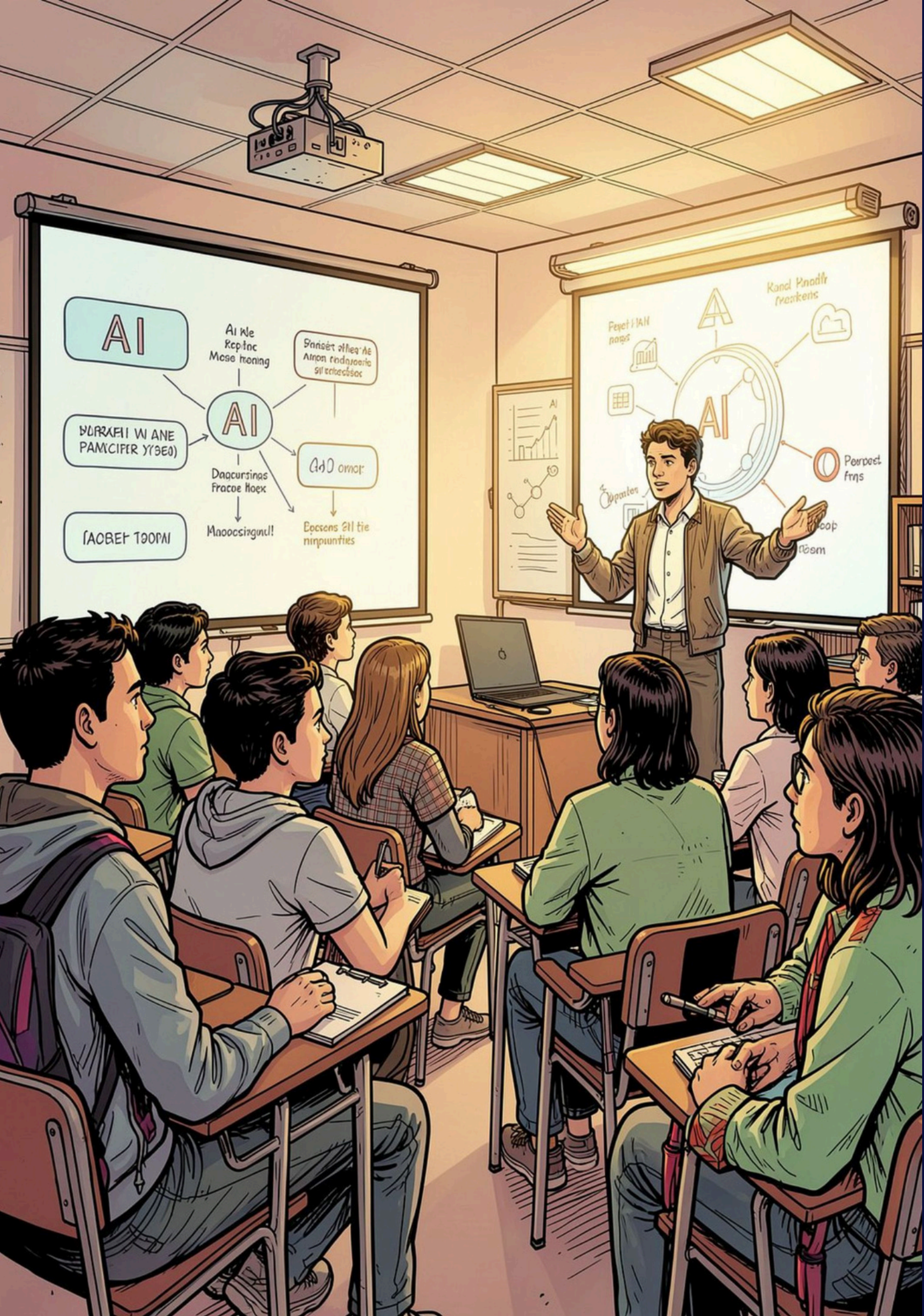
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Roll Number:_26scs1003003658_

College Name : _IILM University_

Internship Duration: 1 Month (Codec Technologies India)





About the Internship

A structured one-month programme at Codec Technologies India combining theory and guided practicals. Key areas covered:

- Artificial Intelligence fundamentals
- Python for data handling
- Machine Learning and Neural Networks
- Deep Learning and advanced AI applications
- Project-oriented learning with prescribed tasks and guided exercises

□ Course mixed lectures, hands-on labs, and supervised project exposure.



Internship Objectives

- Understand the fundamentals of Artificial Intelligence
- Learn Python concepts relevant to AI and data handling
- Understand basic Machine Learning algorithms and workflows
- Explore Neural Networks and Deep Learning ideas
- Recognise practical applications of AI across domains
- Develop technical and analytical problem-solving skills

Internship Roadmap – 4 Weeks

1

Week 1

Foundations of AI & Python for Data Handling

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Week 2

Machine Learning Basics

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Week 3

Neural Networks & Deep Learning

4

Week 4

Advanced AI Applications & Project Development



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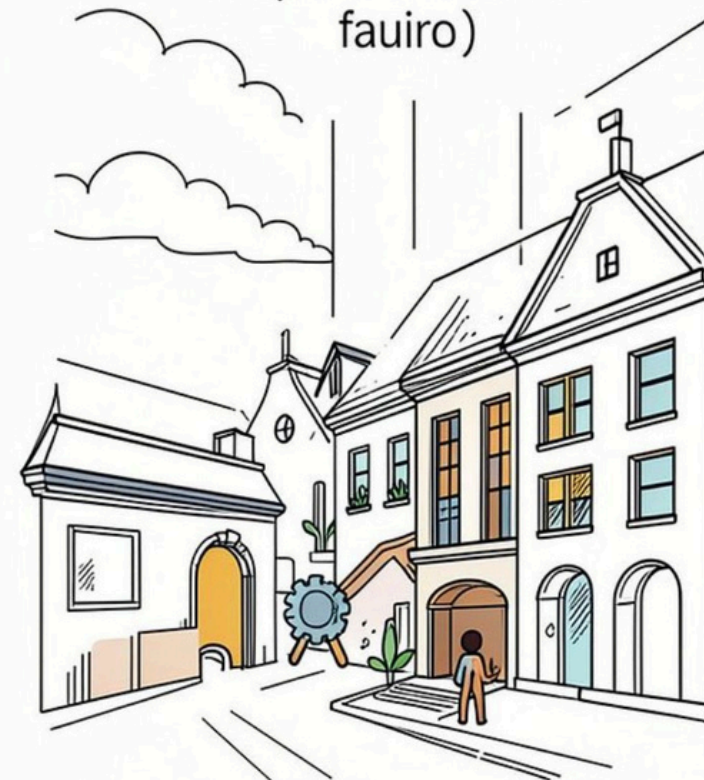


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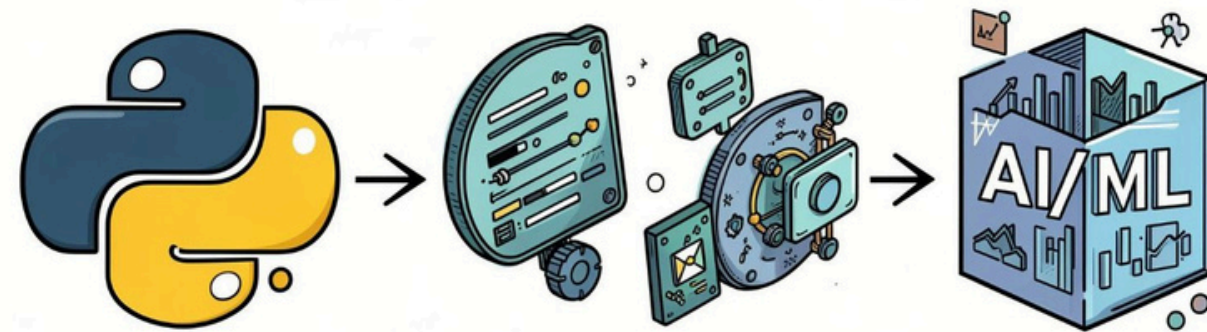


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Week 1 – AI and Python

Focus: core AI ideas and practical Python skills for data work.



- Introduction to Artificial Intelligence and common use-cases
- Basic AI concepts: agents, search, knowledge representation
- Python fundamentals: syntax, data types, control flow
- Data handling: pandas, CSV/JSON, basic cleaning
- Working with datasets and simple exploratory analysis



Visual summary: Python → Data → Processing → AI / ML

Week 2 — Machine Learning

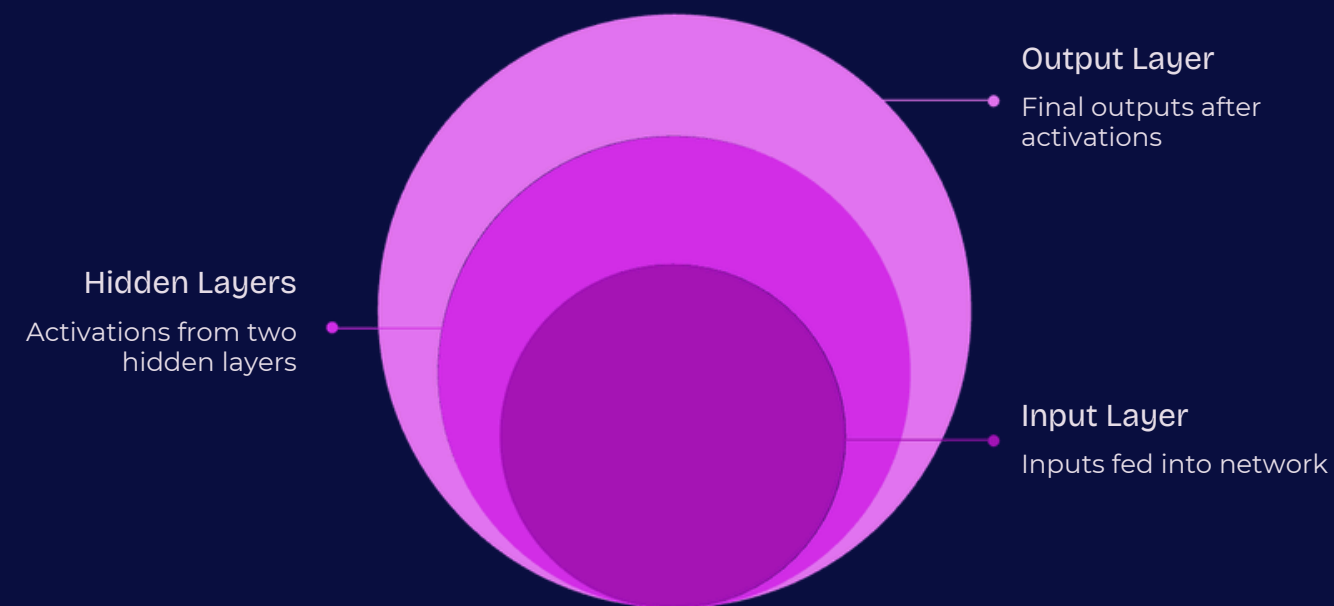
Hands-on introduction to supervised and unsupervised methods and basic model workflows.

- Machine Learning fundamentals and common terminology
- Supervised learning: classification and regression concepts
- Unsupervised learning: clustering and dimensionality reduction
- Basic model training and validation workflows
- Model evaluation metrics and simple interpretation

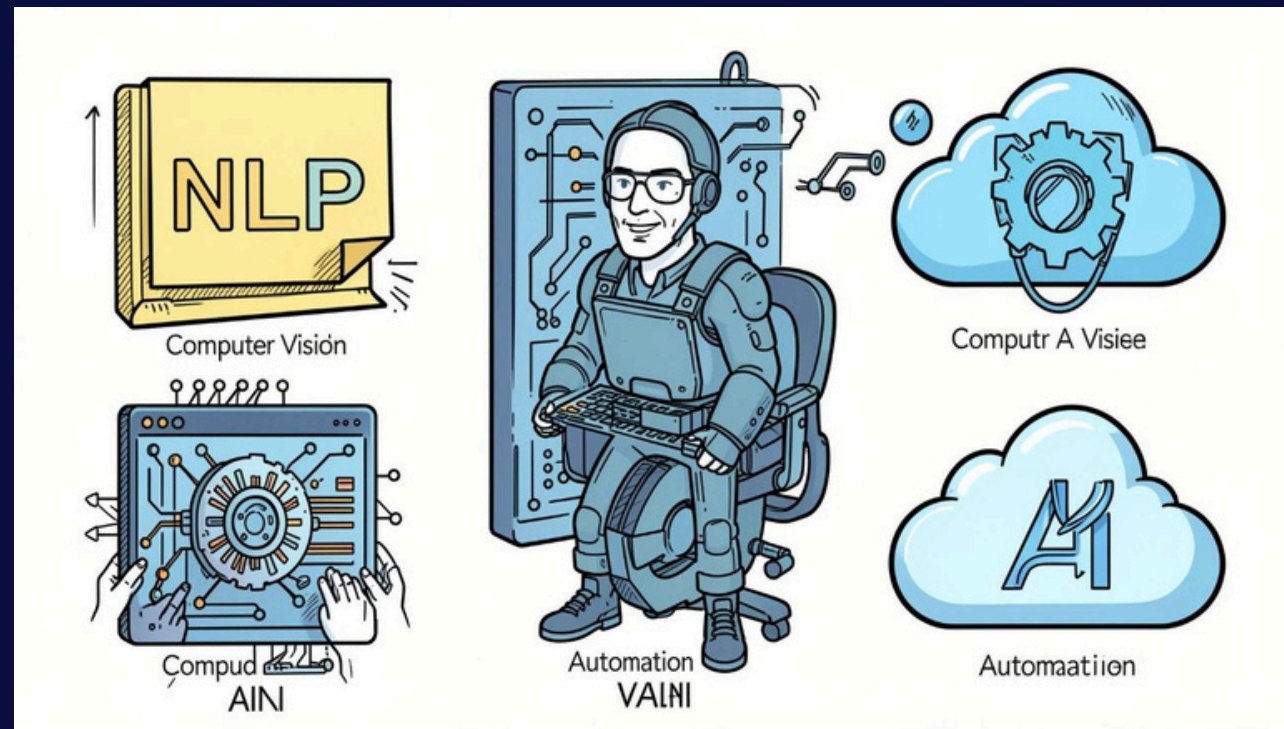


Week 3 — Neural Networks & Deep Learning

Core ideas behind neural models and an introduction to deep learning.



- Artificial neurons and layer-based architectures
- Input, hidden and output layers; forward pass intuition
- Activation functions (ReLU, sigmoid, softmax)
- Training concepts: loss, backpropagation, optimisation basics
- Intro to Deep Learning and real-world applications



Week 4 – Advanced AI Applications

Exposure to higher-level AI topics and project planning concepts.



Natural Language Processing

Text processing, tokenization, basic sentiment tasks.



Computer Vision

Image preprocessing and CNN concepts for classification tasks.



Intelligent Automation

Use-cases for automation with AI-driven decision support.

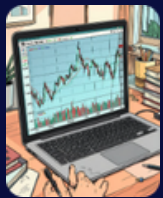


Project Development Concepts

From requirement understanding to evaluation and documentation.

Project Options / Project Exposure

The internship provided a set of prescribed project tasks for practical application of learned concepts. These were offered as options for guided practice or coursework exposure.



Stock Price Predictor

Prediction using historical stock data (project option provided).



Sentiment Analysis on Twitter Data

Classify text as positive, negative or neutral (project option provided).



Handwritten Digit Recognizer

CNN-based digit recognition using MNIST (project option provided).



Chatbot for Customer Service

Rule-based or basic NLP chatbot (project option provided).



Spam Email Classifier

Classify emails as spam or non-spam (project option provided).



Image Classifier for Fruits

Image classification using CNNs (project option provided).



Weather Data Analysis & Prediction

Analyze historical weather data for trends (project option provided).

📌 These are labelled clearly as project options provided by the internship; not completed independently.

AI Concepts & Technologies Explored

Key tools and topics covered during the course of training:

- Python and libraries for data handling (pandas, numpy)
- Data handling and preprocessing
- Machine Learning fundamentals and algorithms
- Neural Networks and Deep Learning concepts
- Natural Language Processing and Computer Vision basics
- Datasets, model training workflows and evaluation



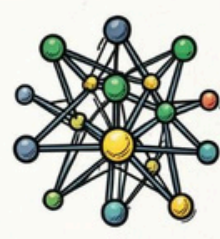
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